

PSOT_2.2, 2.3

Department of Electronics and Telecommunications

SUMMARIZING DATA SETS

- **Arithmetic Mean, Geometric Mean, Harmonic Mean**
- **Median, Mode, Variance, Standard Deviation, Percentile**
- **Correlation, Covariance**

Arithmetic Mean

- Suppose that we have a data set consisting of the n numerical values $x_1, x_2, \dots, x_n$
- The sample mean is the arithmetic average of these values

The *sample mean*, designated by $\bar{x}$, is defined by

$$\bar{x} = \sum_{i=1}^n x_i / n$$

EXAMPLE 2.3b The following is a frequency table giving the ages of members of a symphony orchestra for young adults.

Age	Frequency
15	2
16	5
17	11
18	9
19	14
20	13

$$AM = \frac{\sum f \cdot x}{\sum f}$$

Find the sample mean of the ages of the 54 members of the symphony.

SOLUTION

$$\bar{x} = (15 \cdot 2 + 16 \cdot 5 + 17 \cdot 11 + 18 \cdot 9 + 19 \cdot 14 + 20 \cdot 13)/54 \approx 18.24$$

The formula for the **Geometric Mean (GM)** of n positive numbers $x_1, x_2, \dots, x_n$ is:

$$GM = \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$$

For Grouped Data (with frequencies):

If the data has corresponding frequencies $f_1, f_2, \dots, f_n$, the formula becomes:

$$GM = \sqrt[\sum f]{x_1^{f_1} \cdot x_2^{f_2} \cdot \dots \cdot x_n^{f_n}}$$

This is often expressed logarithmically to simplify multiplication:

$$GM = \exp \left(\frac{\sum f \cdot \log(x)}{\sum f} \right)$$

The formula for the **Harmonic Mean (HM)** of n positive numbers $x_1, x_2, \dots, x_n$ is:

$$HM = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

For Grouped Data (with Frequencies):

If the data has corresponding frequencies $f_1, f_2, \dots, f_n$, the formula becomes:

$$HM = \frac{\sum f}{\sum \frac{f}{x}}$$

EXAMPLE 2.3b The following is a frequency table giving the ages of members of a symphony orchestra for young adults.

Age	Frequency
15	2
16	5
17	11
18	9
19	14
20	13

Find Geometric Mean and Harmonic Mean

Age (x)	Frequency (f)	$f \cdot x$	$f \cdot \log(x)$	$\frac{f}{x}$
15	2	$15 \cdot 2 = 30$	$2 \cdot \log(15)$	$\frac{2}{15} = 0.1333$
16	5	$16 \cdot 5 = 80$	$5 \cdot \log(16)$	$\frac{5}{16} = 0.3125$
17	11	$17 \cdot 11 = 187$	$11 \cdot \log(17)$	$\frac{11}{17} = 0.6471$
18	9	$18 \cdot 9 = 162$	$9 \cdot \log(18)$	$\frac{9}{18} = 0.5000$
19	14	$19 \cdot 14 = 266$	$14 \cdot \log(19)$	$\frac{14}{19} = 0.7368$
20	13	$20 \cdot 13 = 260$	$13 \cdot \log(20)$	$\frac{13}{20} = 0.6500$
Total Frequency (N) = $2 + 5 + 11 + 9 + 14 + 13 = 54$				

EXAMPLE 2.3b The following is a frequency table giving the ages of members of a symphony orchestra for young adults.

Age	Frequency
15	2
16	5
17	11
18	9
19	14
20	13

Find Arithmetic mean, Geometric Mean and Harmonic Mean

Arithmetic Mean (AM): 18.24

Geometric Mean (GM): 18.08

Harmonic Mean (HM): 18.12

When to Use Each:

Mean Type	Use When	Example
Arithmetic Mean	Data is linearly distributed, without extreme outliers.	Average sales of a store over 12 months.
Geometric Mean	Data involves growth rates, percentages, or proportions.	Average annual growth rate of a company's revenue over 5 years.
Harmonic Mean	Data involves rates (e.g., speed, efficiency, or frequency) or minimizing large outlier effects.	Calculating average speed when traveling at different rates over distances.

Sample Median

- It is the middle value when the data set is arranged in increasing order
- Procedure to find median
 - Order the values of a data set of size n from smallest to largest
 - if n is even, it is the average of the values in positions $n/2$ and $n/2+1$
 - If n is odd, the *sample median* is the value in position $(n+1)/2$

Germ-Free Mice

1	58, 92, 93, 94, 95
2	02, 12, 15, 29, 30, 37, 40, 44, 47, 59
3	01, 01, 21, 37
4	15, 34, 44, 85, 96
5	29, 37
6	24
7	07
8	00

Conventional Mice

1	59, 89, 91, 98
2	35, 45, 50, 56, 61, 65, 66, 80
3	43, 56, 83
4	03, 14, 28, 32

Determine the sample means and the sample medians for the two sets of mice.

Solution

- Germ-Free Mice sample mean = 344.07,
- Conventional Mice sample mean = 292.32
- The Germ-Free Mice sample median is the 15th largest data value, namely, 259
- Sample median for the Conventional Mice is the 10th largest data value, namely, 265

Sample mode

- Defined to be the value that occurs with the greatest frequency

EXAMPLE 2.3e The following frequency table gives the values obtained in 40 rolls of a die.

Value	Frequency
1	9
2	8
3	5
4	5
5	6
6	7

Find (a) the sample mean, (b) the sample median, and (c) the sample mode.

Solution

(a) The sample mean is

$$\bar{x} = (9 + 16 + 15 + 20 + 30 + 42)/40 = 3.05$$

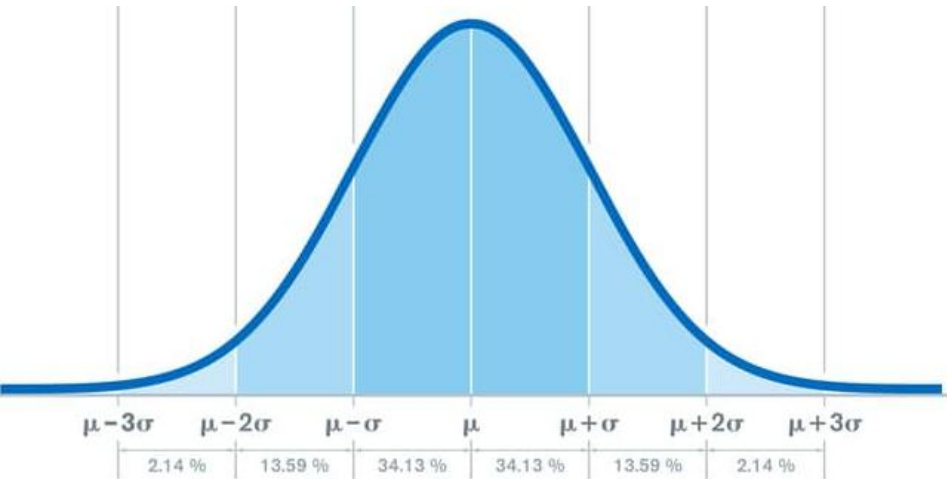
(b) The sample median is the average of the 20th and 21st smallest values,
equal to 3.

(c) The sample mode is 1, the value that occurred most frequently.

Sample Variance

- Describe the spread or variability of the data values

The *sample variance*, call it s^2 , of the data set $x_1, \dots, x_n$ is defined by



$$s^2 = \sum_{i=1}^n (x_i - \bar{x})^2 / (n - 1)$$

EXAMPLE 2.3f Find the sample variances of the data sets **A** and **B** given below.

$$\mathbf{A}: 3, 4, 6, 7, 10 \quad \mathbf{B}: -20, 5, 15, 24$$

SOLUTION As the sample mean for data set **A** is $\bar{x} = (3 + 4 + 6 + 7 + 10)/5 = 6$, it follows that its sample variance is

$$s^2 = [(-3)^2 + (-2)^2 + 0^2 + 1^2 + 4^2]/4 = 7.5$$

The sample mean for data set **B** is also 6; its sample variance is

$$s^2 = [(-26)^2 + (-1)^2 + 9^2 + (18)^2]/3 \approx 360.67$$

EXAMPLE 2.3g The following data give the worldwide number of fatal airline accidents of commercially scheduled air transports in the years from 1997 to 2005.

Year	1997	1998	1999	2000	2001	2002	2003	2004	2005
Accidents	25	20	21	18	13	13	7	9	18

Source: National Safety Council.

Find the sample variance of the number of accidents in these years.

Ans : 34.75

Sample Standard Deviation

The quantity s , defined by

$$s = \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 / (n - 1)}$$

is called the *sample standard deviation*.

Sample Percentiles

- The *sample 100p percentile* is that data value such that 100p percent of the data are less than or equal to it and 100(1-p) percent are greater than or equal to it.
- The sample 25th percentile is called the *first quartile*
- The sample 50th percentile is called the sample median or the *second quartile*
- The sample 75th percentile is called the *third quartile*.

Sample Percentiles

- To accomplish this:
 - First arrange the data in increasing order
 - If n is number of samples and p is the percentile to be calculated, then
 - if np is not an integer, then sample percentile is the smallest value exceeding np
 - if np is an integer, then both the values in positions np and $np+1$ satisfy the preceding conditions, and so the sample $100p$ percentile is the average of these values.

Avg. Temperature of cities

7	0.0
6	9.0
5	1.0, 1.3, 2.0, 5.5, 7.1, 7.4, 7.6, 8.5, 9.3
4	0.0, 1.0, 2.4, 3.6, 3.7, 4.8, 5.0, 5.2, 6.0, 6.7, 8.1, 9.0, 9.2
3	3.1, 4.1, 5.3, 5.8, 6.2, 9.0, 9.5, 9.5
2	9.0, 9.8 ■

- Find sample 10th percentile
- $34 * 0.1 = 3.4$, so 4th smallest value – 34.1

TABLE 2.6 *Population of 25 Largest U.S. Cities, July 2006*

Rank	City	Population
1	New York, NY	8,250,567
2	Los Angeles, CA	3,849,378
3	Chicago, IL	2,833,321
4	Houston, TX	2,144,491
5	Phoenix, AR	1,512,986
6	Philadelphia, PA	1,448,394
7	San Antonio, TX	1,296,682
8	San Diego, CA	1,256,951
9	Dallas, TX	1,232,940
10	San Jose, CA	929,936
11	Detroit, MI	918,849
12	Jacksonville, FL	794,555
13	Indianapolis, IN	785,597
14	San Francisco, CA	744,041
15	Columbus, OH	733,203
16	Austin, TX	709,893
17	Memphis, TN	670,902
18	Fort Worth, TX	653,320
19	Baltimore, MD	640,961
20	Charlotte, NC	630,478
21	El Paso, TX	609,415
22	Milwaukee, WI	602,782
23	Boston, MA	590,763
24	Seattle, WA	582,454
25	Washington, DC	581,530

Find Sample 10 percentile

Find Sample 80 Percentile
for the given data

Solution:

SOLUTION (a) Because the sample size is 25 and $25(.10) = 2.5$, the sample 10 percentile is the third smallest value, equal to 590,763.

(b) Because $25(.80) = 20$, the sample 80 percentile is the average of the twentieth and the twenty-first smallest values. Hence, the sample 80 percentile is

$$\frac{1,512,986 + 1,448,394}{2} = 1,480,690 \quad \blacksquare$$

The following data give noise levels measured at 36 different times directly outside of Grand Central Station in Manhattan.

82, 89, 94, 110, 74, 122, 112, 95, 100, 78, 65, 60, 90, 83, 87, 75, 114, 85
69, 94, 124, 115, 107, 88, 97, 74, 72, 68, 83, 91, 90, 102, 77, 125, 108, 65

Determine the quartiles.

SOLUTION A stem and leaf plot of the data is as follows:

6	0, 5, 5, 8, 9
7	2, 4, 4, 5, 7, 8
8	2, 3, 3, 5, 7, 8, 9
9	0, 0, 1, 4, 4, 5, 7
10	0, 2, 7, 8
11	0, 2, 4, 5
12	2, 4, 5

SOLUTION A stem and leaf plot of the data is as follows:

6	0, 5, 5, 8, 9
7	2, 4, 4, 5, 7, 8
8	2, 3, 3, 5, 7, 8, 9
9	0, 0, 1, 4, 4, 5, 7
10	0, 2, 7, 8
11	0, 2, 4, 5
12	2, 4, 5

- First quartile: $36 \times .25 = 9$, avg of 9th and 10th smallest value = 76
- Second quartile: $36 \times .5 = 18$, avg of 18th and 19th smallest value = 89.5
- Third quartile: $36 \times .75 = 27$, avg of 27th and 28th smallest value = 104.5

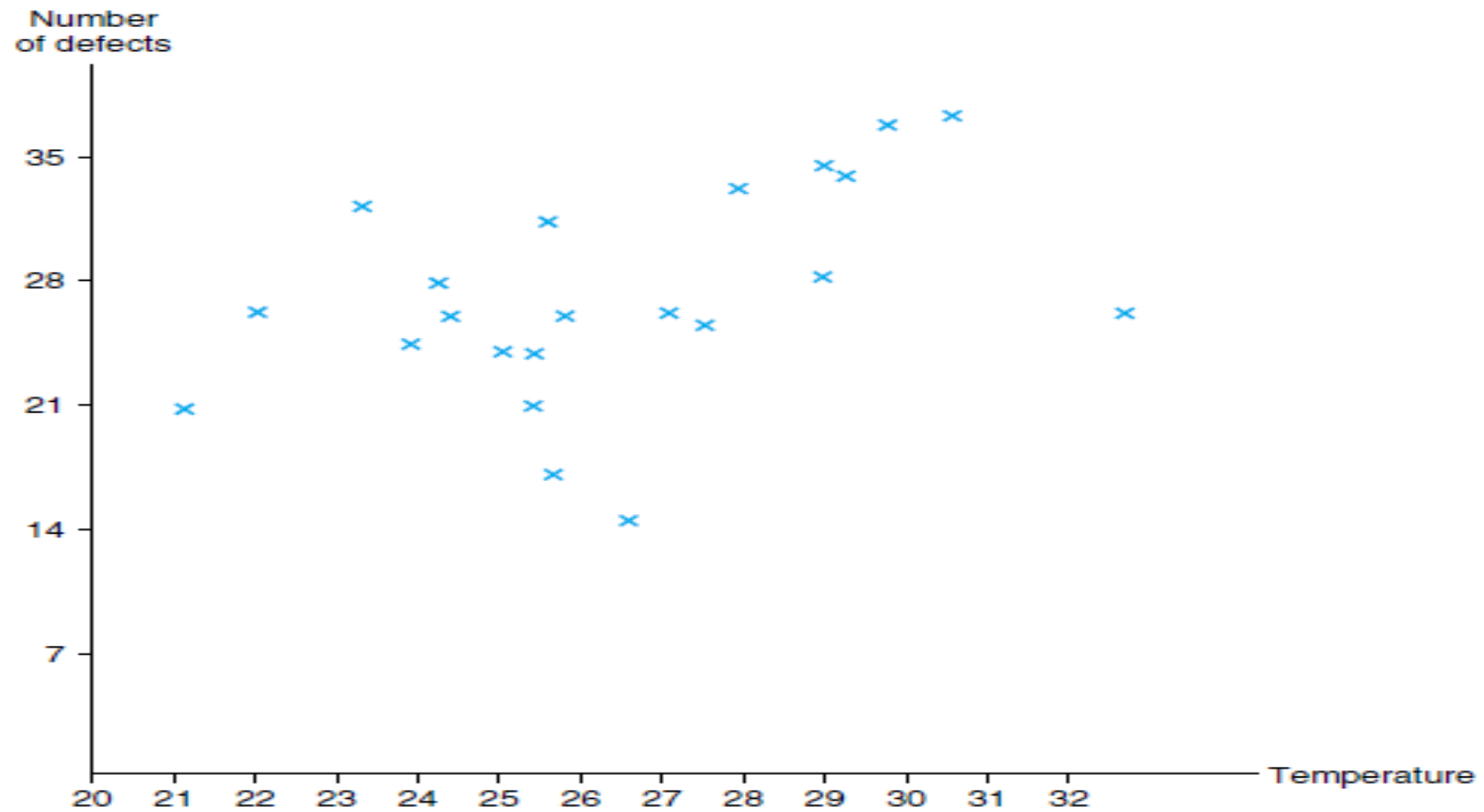
PAIRED DATA SETS

- Sets that consist of pairs of values that have some relationship to each other
- Data set of paired values is to plot the data on a two dimensional graph,
- The x-axis representing the x value of the data
- The y -axis representing the y value. Such a plot is called a *scatter diagram*

TABLE 2.8 *Temperature and Defect Data*

Day	Temperature	Number of Defects
1	24.2	25
2	22.7	31
3	30.5	36
4	28.6	33
5	25.5	19
6	32.0	24
7	28.6	27
8	26.5	25
9	25.3	16
10	26.0	14
11	24.4	22
12	24.8	23
13	20.6	20
14	25.1	25
15	21.4	25
16	23.7	23
17	23.9	27
18	25.2	30
19	27.4	33
20	28.3	32
21	28.8	35
22	26.6	24

Scatter Diagram



Sample correlation coefficient

Let s_x and s_y denote, respectively, the sample standard deviations of the x values and the y values. The *sample correlation coefficient*, call it r , of the data pairs (x_i, y_i) , $i = 1, \dots, n$ is defined by

$$\begin{aligned} r &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \end{aligned}$$

Properties of r

1. $-1 \leq r \leq 1$
2. If for constants a and b , with $b > 0$,

$$y_i = a + bx_i, \quad i = 1, \dots, n$$

then $r = 1$.

3. If for constants a and b , with $b < 0$,

$$y_i = a + bx_i, \quad i = 1, \dots, n$$

then $r = -1$.

4. If r is the sample correlation coefficient for the data pairs $x_i, y_i, i = 1, \dots, n$ then it is also the sample correlation coefficient for the data pairs

$$a + bx_i, \quad c + dy_i, \quad i = 1, \dots, n$$

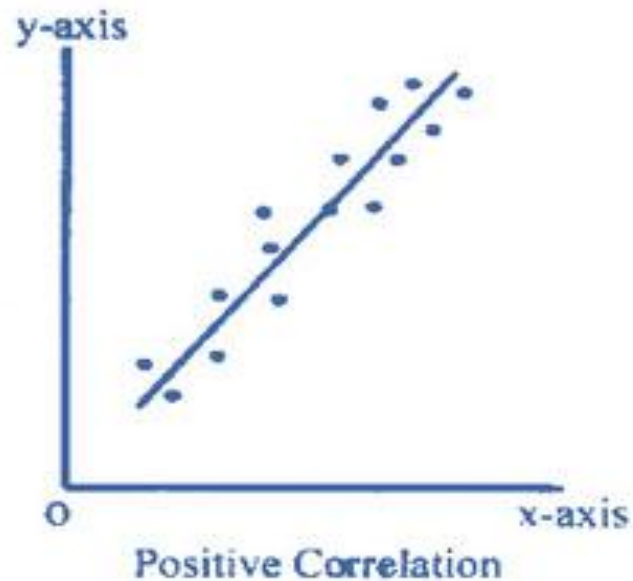
provided that b and d are both positive or both negative.

Sample correlation coefficient

- The absolute value of the sample correlation coefficient r (that is, $|r|$, its value without regard to its sign) is a measure of the strength of the linear relationship between the x and the y values of a data pair
- A value of $|r|$ equal to 1 means that there is a perfect linear relation

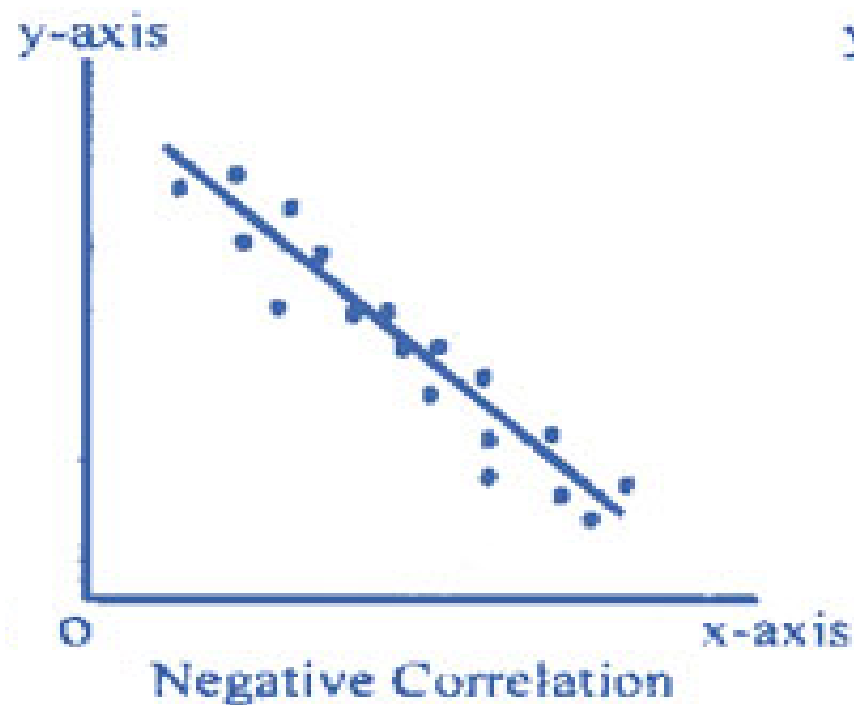
Sample correlation coefficient

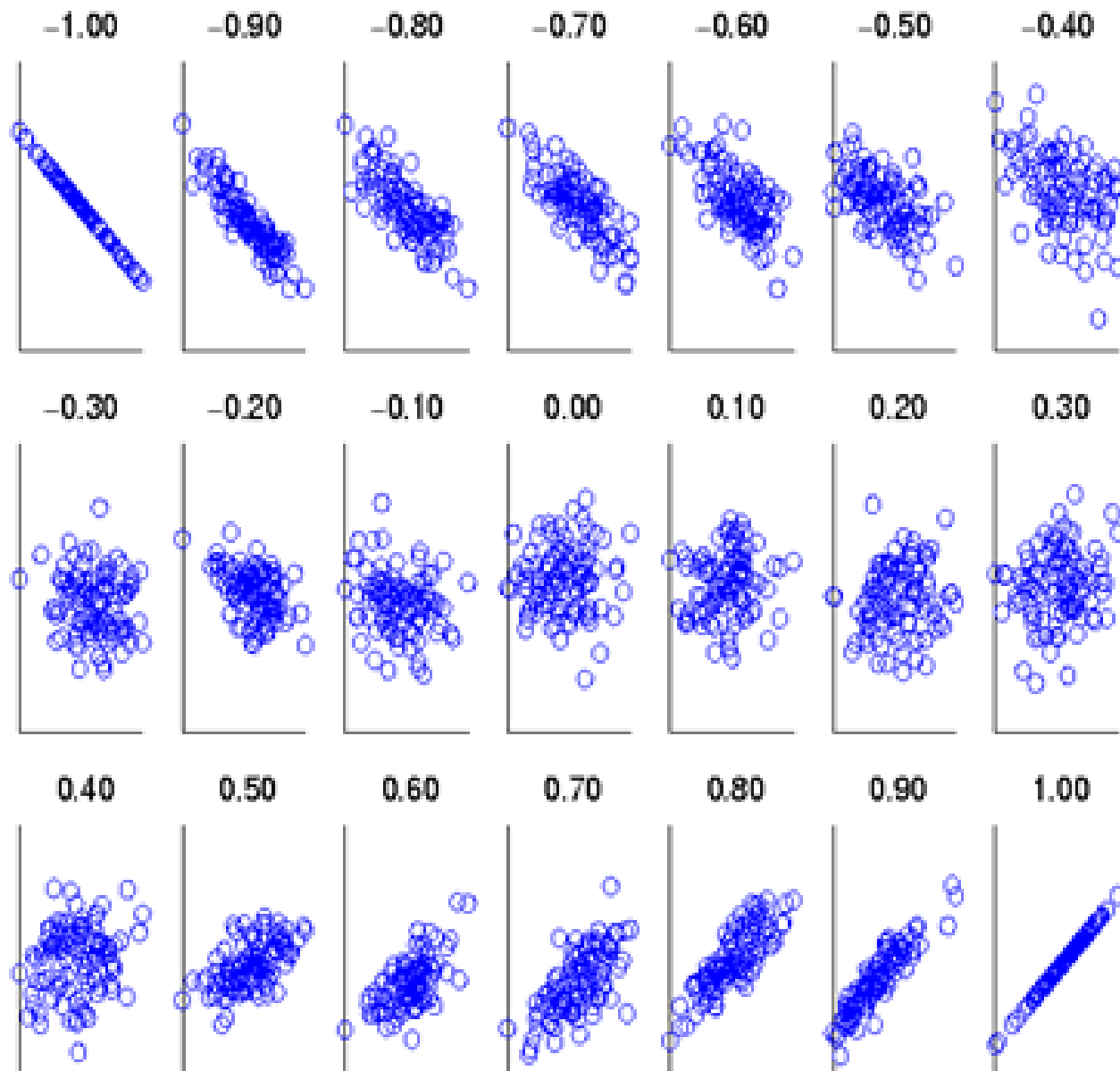
- The sign of r gives the direction of the relation
- r is positive when the linear relation is such that smaller y values tend to go with smaller x values and larger y values with larger x values



Sample correlation coefficient

- r is negative when larger y values tend to go with smaller x values and smaller y values with larger x values





Covariance

- It measures the direction of the linear relationship between two variables.
- It is a foundational concept used to understand relationships in data and informs decision-making in various areas of data analytics.

Applications of Covariance:

1. Understanding Relationships Between Variables

Sales vs. Advertising Spend: A positive covariance suggests that increasing advertising spend leads to higher sales.

2. Input for Correlation

While covariance indicates the direction of the relationship, it does not standardize the scale. Covariance is often used to compute the **correlation coefficient**, which provides a normalized measure of the strength and direction of a relationship.

$$\text{Correlation} = \frac{\text{Covariance}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Covariance

3. Feature Selection in Machine Learning

Covariance is used in **Principal Component Analysis (PCA)** to reduce dimensionality.

The covariance matrix captures the relationship between multiple features. PCA identifies directions (principal components) with the highest variance, simplifying the dataset while retaining essential information.

Example: Reducing the number of variables in a dataset while retaining its predictive power.

EXAMPLE 2.3f Find the sample variances of the data sets **A** and **B** given below.

A: 3, 4, 6, 7, 10 **B:** -20, 5, 15, 24

Find Covariance

$$\text{Covariance}(X, Y) = \frac{\sum ((X_i - \bar{X})(Y_i - \bar{Y}))}{n - 1}$$

EXAMPLE 2.3f Find the sample variances of the data sets **A** and **B** given below.

A: 3, 4, 6, 7, 10 **B**: -20, 5, 15, 24

$$\text{Covariance}(X, Y) = \frac{\sum ((X_i - \bar{X})(Y_i - \bar{Y}))}{n - 1}$$

We calculate the deviations for both datasets and their product:

X_i	$X_i - \bar{X}$	Y_i	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$
3	$3 - 6 = -3$	-20	$-20 - 6 = -26$	$(-3)(-26) = 78$
4	$4 - 6 = -2$	5	$5 - 6 = -1$	$(-2)(-1) = 2$
6	$6 - 6 = 0$	15	$15 - 6 = 9$	$(0)(9) = 0$
7	$7 - 6 = 1$	24	$24 - 6 = 18$	$(1)(18) = 18$
10	$10 - 6 = 4$			

$$\text{Covariance}(X, Y) = \frac{78 + 2 + 0 + 18}{4 - 1} = \frac{98}{3} = 32.67$$

EXAMPLE 2.3f Find the sample variances of the data sets **A** and **B** given below.

$$\mathbf{A}: 3, 4, 6, 7, 10 \quad \mathbf{B}: -20, 5, 15, 24$$

SOLUTION As the sample mean for data set **A** is $\bar{x} = (3 + 4 + 6 + 7 + 10)/5 = 6$, it follows that its sample variance is

$$s^2 = [(-3)^2 + (-2)^2 + 0^2 + 1^2 + 4^2]/4 = 7.5$$

The sample mean for data set **B** is also 6; its sample variance is

$$s^2 = [(-26)^2 + (-1)^2 + 9^2 + (18)^2]/3 \approx 360.67$$

Find Sample correlation coefficient, $r = 0.63$

$$\begin{aligned} r &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \end{aligned}$$

• $r = .4189$

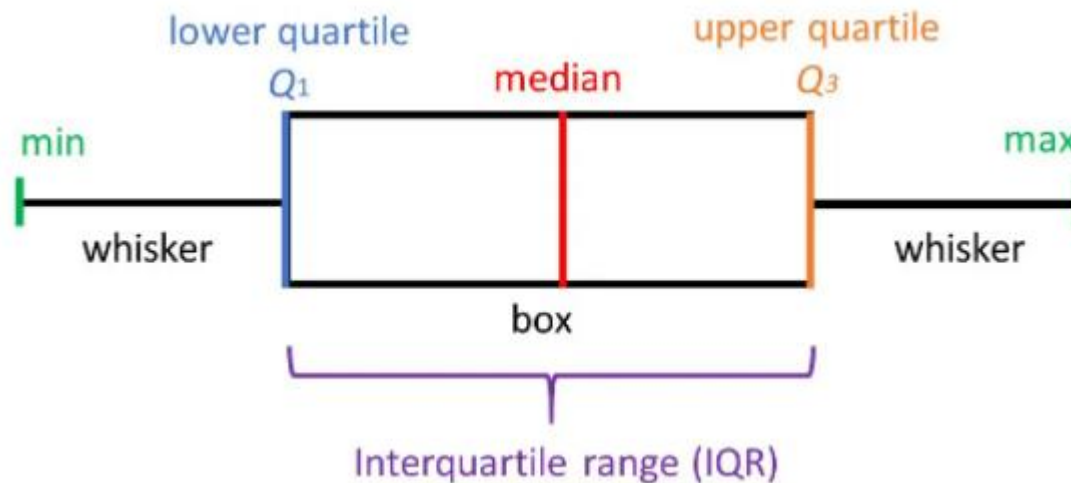
TABLE 2.8 *Temperature and Defect Data*

Day	Temperature	Number of Defects
1	24.2	25
2	22.7	31
3	30.5	36
4	28.6	33
5	25.5	19
6	32.0	24
7	28.6	27
8	26.5	25
9	25.3	16
10	26.0	14
11	24.4	22
12	24.8	23
13	20.6	20
14	25.1	25
15	21.4	25
16	23.7	23
17	23.9	27
18	25.2	30
19	27.4	33
20	28.3	32
21	28.8	35
22	26.6	24



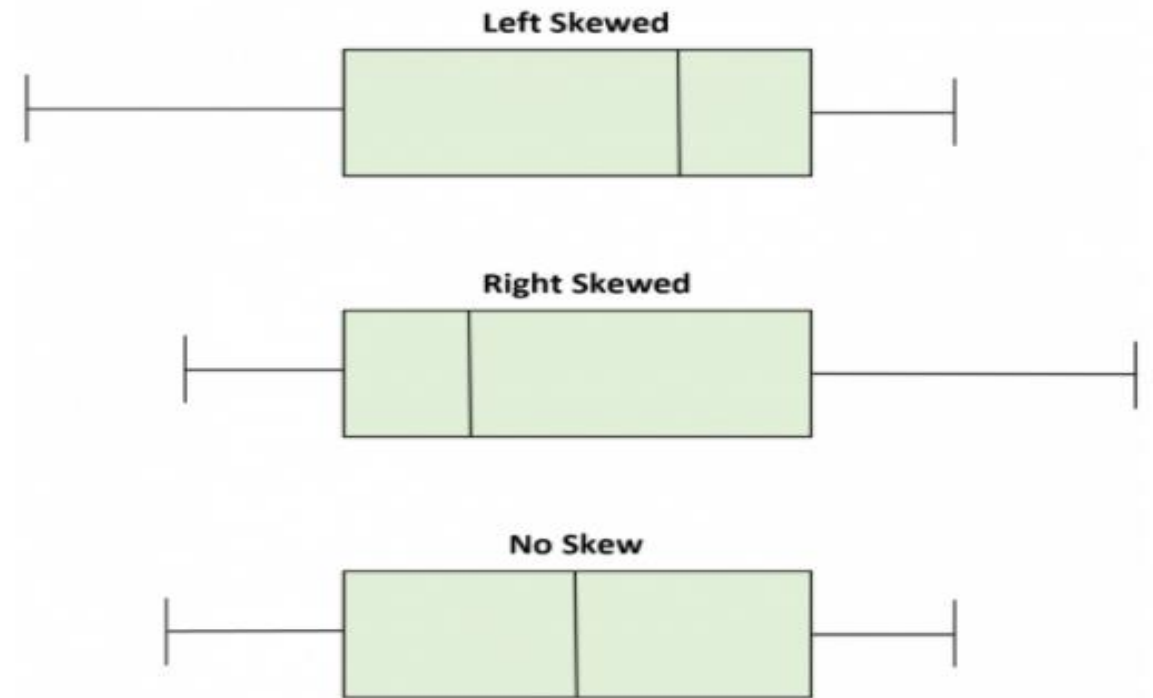
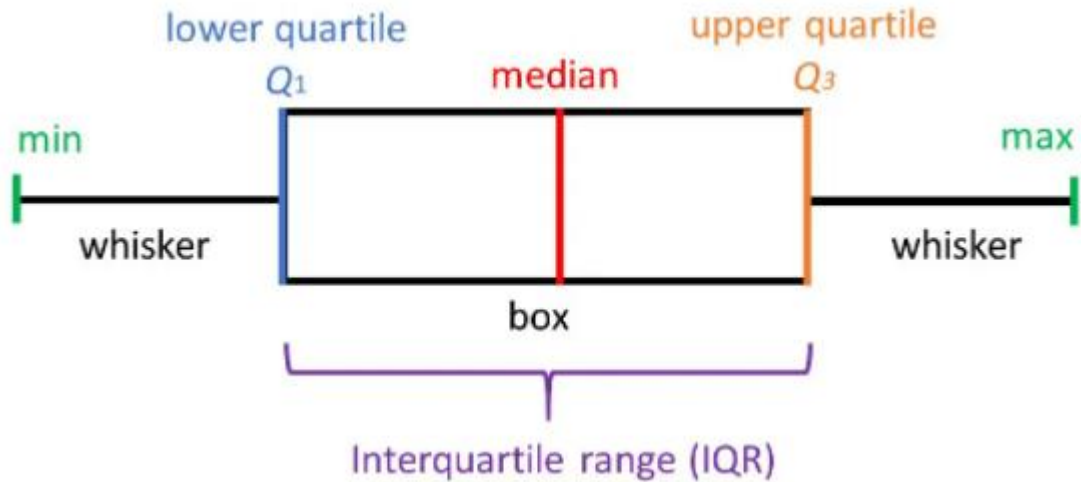
Box plot

- A boxplot is a graph that gives you a good indication of **how the values in the data are spread out**
- A straight line segment from the smallest to the largest data value is drawn on a horizontal axis
- Imposed on the line is a “box,” which starts at the first and continues to the third quartile, with the value of the second quartile indicated by a vertical line



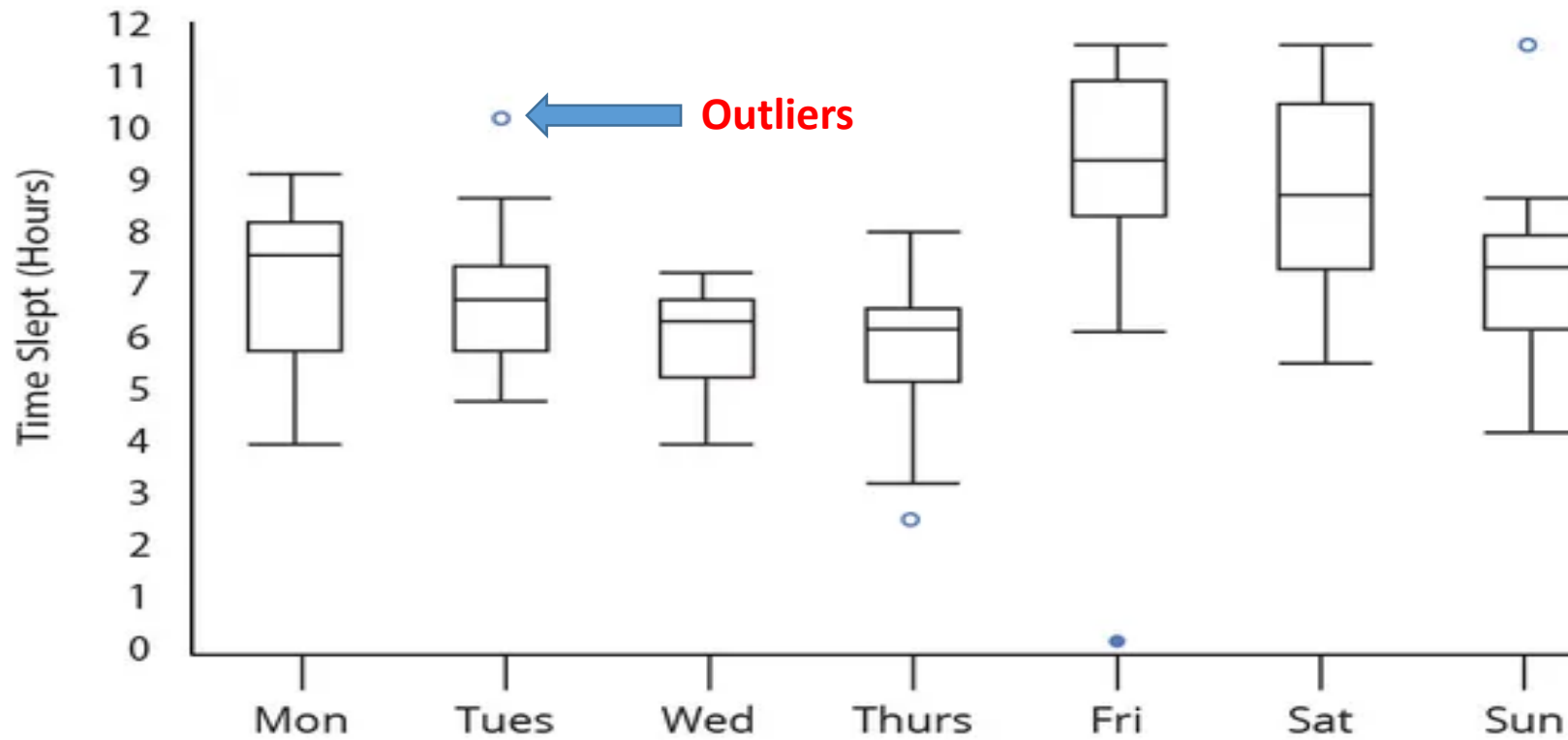
Box plot

- The **upper and lower whiskers** represent scores outside the lower 25% of scores and the upper 25% of scores.
- Box plots are useful as they show the **skewness of a data set**



Box plot

- Box plots are useful to Look for potential outliers



Box plot

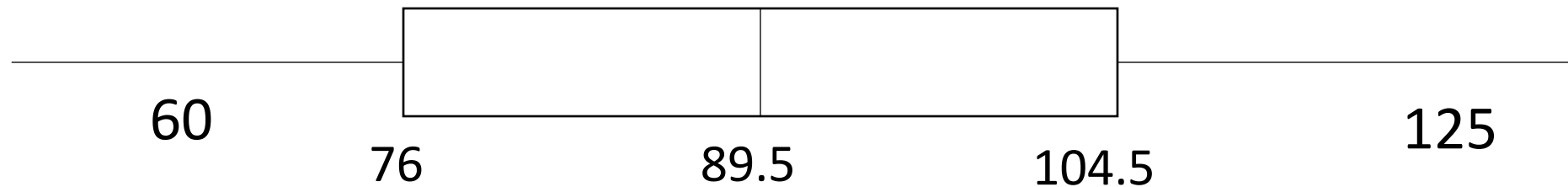
SOLUTION A stem and leaf plot of the data is as follows:

6	0, 5, 5, 8, 9
7	2, 4, 4, 5, 7, 8
8	2, 3, 3, 5, 7, 8, 9
9	0, 0, 1, 4, 4, 5, 7
10	0, 2, 7, 8
11	0, 2, 4, 5
12	2, 4, 5

Box plot

SOLUTION A stem and leaf plot of the data is as follows:

6	0, 5, 5, 8, 9
7	2, 4, 4, 5, 7, 8
8	2, 3, 3, 5, 7, 8, 9
9	0, 0, 1, 4, 4, 5, 7
10	0, 2, 7, 8
11	0, 2, 4, 5
12	2, 4, 5



Data description

- The numerical findings of a study should be presented **clearly, concisely, and in such a manner that an observer can quickly obtain a feel for the essential characteristics of the data.**
- Over the years it has been found that **tables and graphs** are particularly useful ways of presenting data, often revealing important features such as the range, the degree of concentration, and the symmetry of the data.
- Now we will discuss some common graphical and tabular ways for presenting data.

DESCRIBING DATA SETS

- **Frequency Tables and Graphs**
- **Relative Frequency Tables and Graphs**
- **Grouped Data**
- **Histograms**
- **Ogives**
- **Stem and Leaf Plots**



Frequency Tables and Graphs

- Frequency table for a data set consisting of the starting yearly salaries (to the nearest thousand dollars) of 42 recently graduated students

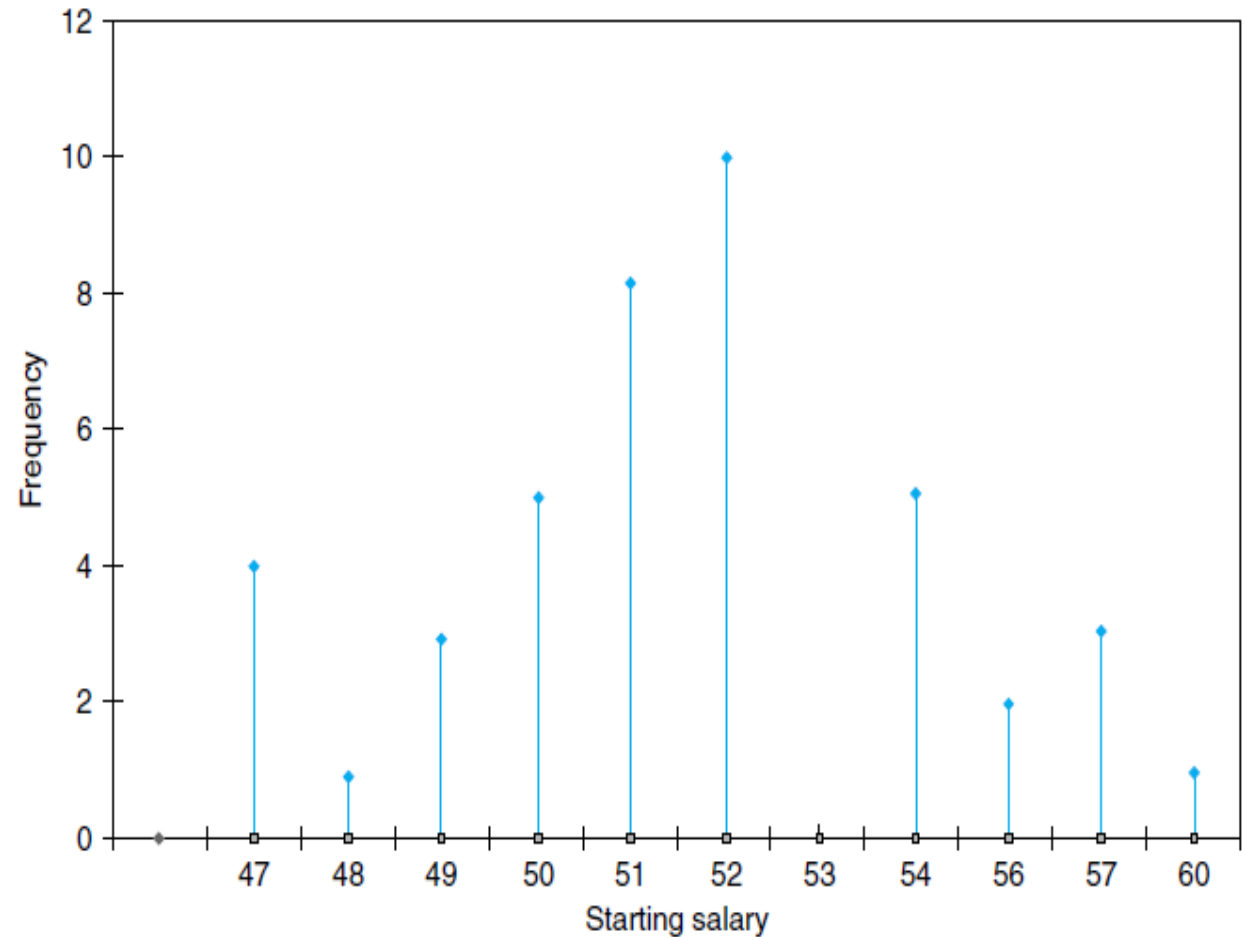
TABLE 2.1 *Starting Yearly Salaries*

Starting Salary	Frequency
47	4
48	1
49	3
50	5
51	8
52	10
53	0
54	5
56	2
57	3
60	1

Line graph

TABLE 2.1 Starting Yearly Salaries

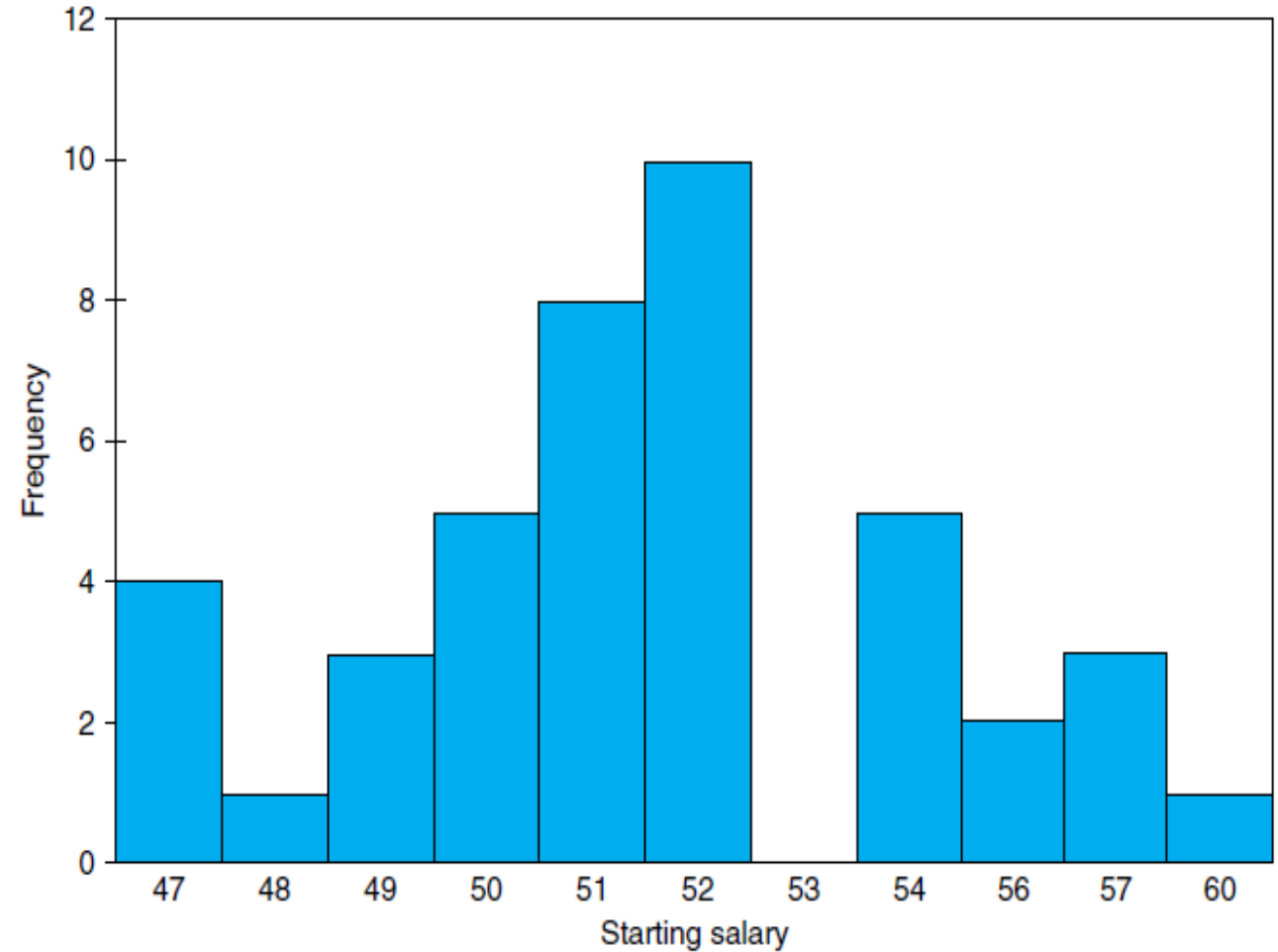
Starting Salary	Frequency
47	4
48	1
49	3
50	5
51	8
52	10
53	0
54	5
56	2
57	3
60	1



Bar graph

TABLE 2.1 *Starting Yearly Salaries*

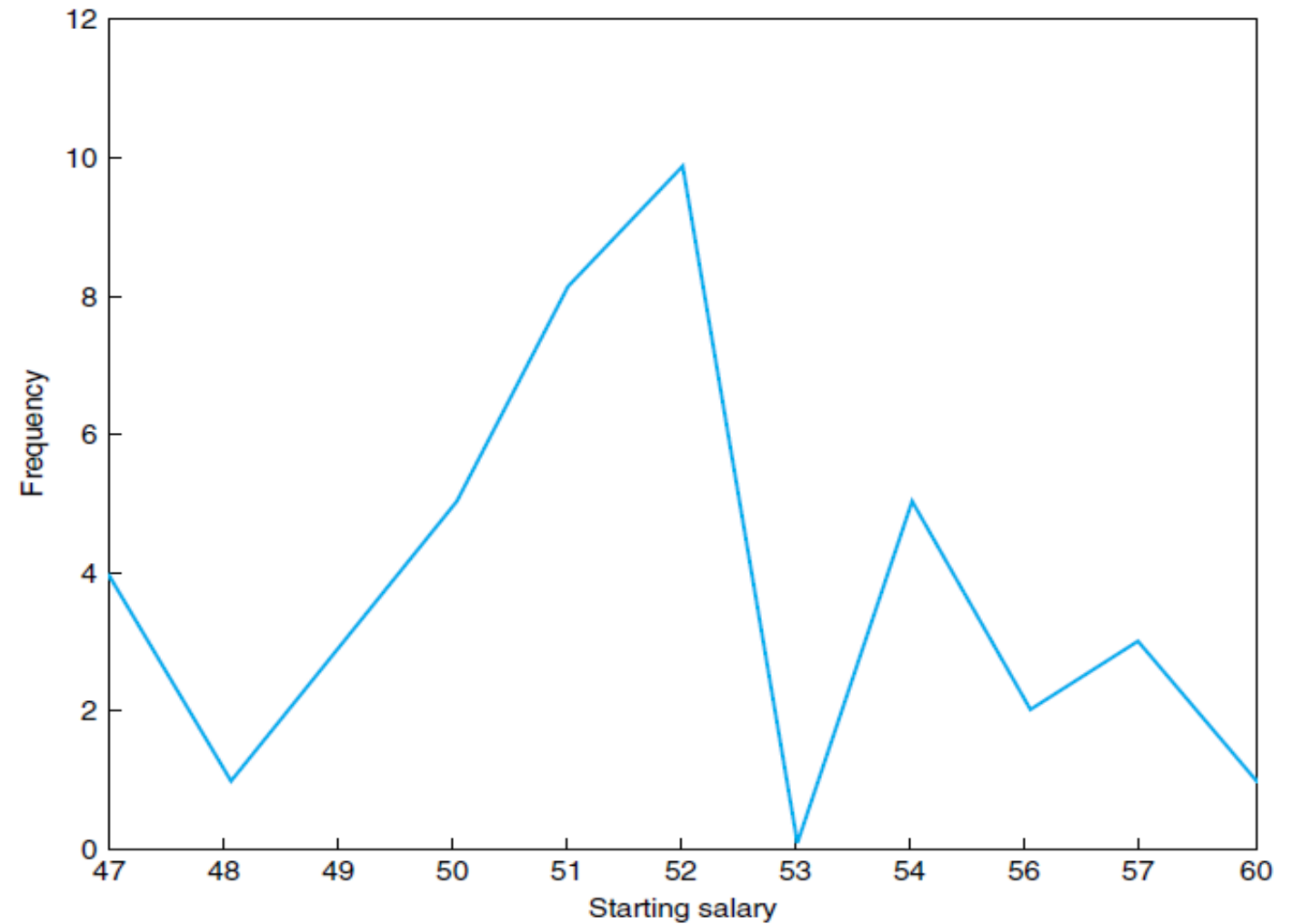
Starting Salary	Frequency
47	4
48	1
49	3
50	5
51	8
52	10
53	0
54	5
56	2
57	3
60	1



Frequency polygon

TABLE 2.1 *Starting Yearly Salaries*

Starting Salary	Frequency
47	4
48	1
49	3
50	5
51	8
52	10
53	0
54	5
56	2
57	3
60	1



Relative Frequency Tables and Graphs

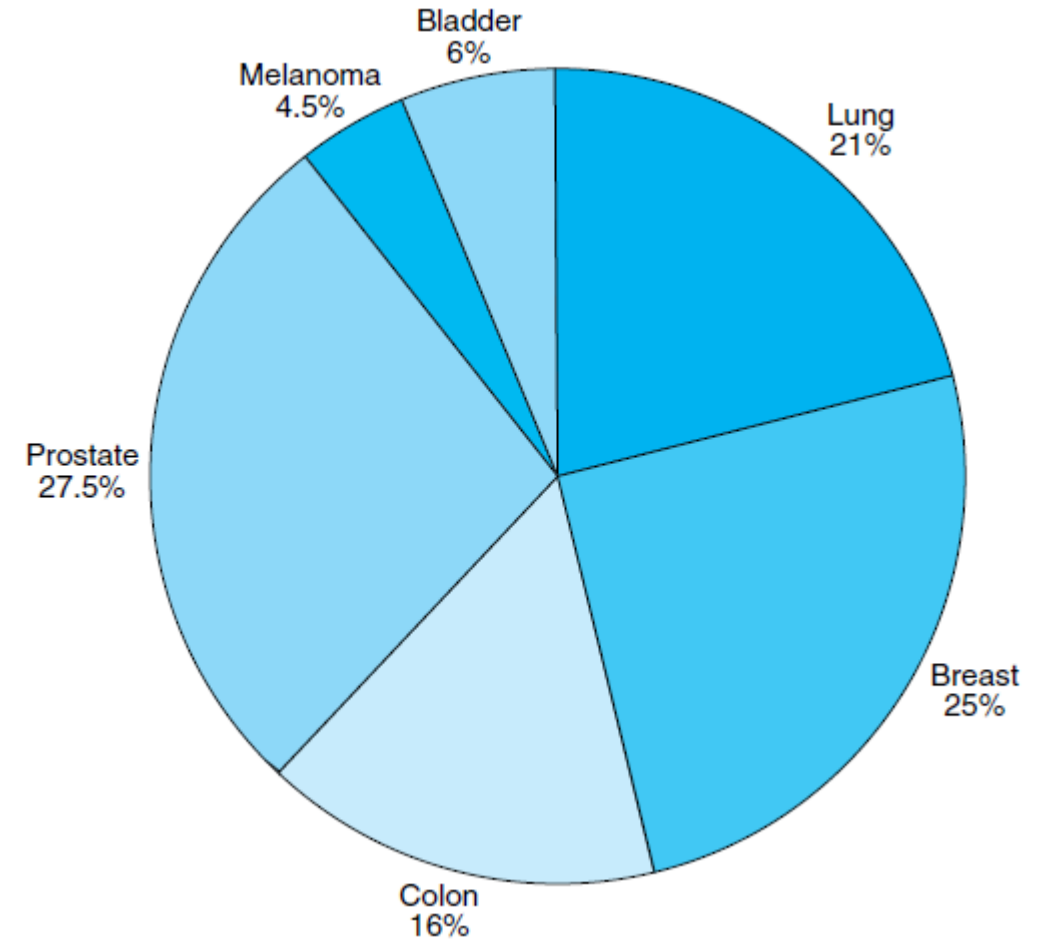
- Consider a data set consisting of n values. If f is the frequency of a particular value, then the ratio f/n is called its *relative frequency*

TABLE 2.2

Starting Salary	Frequency
47	$4/42 = .0952$
48	$1/42 = .0238$
49	$3/42$
50	$5/42$
51	$8/42$
52	$10/42$
53	0
54	$5/42$
56	$2/42$
57	$3/42$
60	$1/42$

Pie Chart

Type of Cancer	Number of New Cases	Relative Frequency
Lung	42	.21
Breast	50	.25
Colon	32	.16
Prostate	55	.275
Melanoma	9	.045
Bladder	12	.06



Grouped Data, Histograms, Ogives, and Stem and Leaf Plots

- Divide the values into *groupings, or class intervals*, and then plot the number of data values falling in each class interval
- The endpoints of a class interval are called the *class boundaries*
- We will adopt the *leftend inclusion convention*

TABLE 2.3 *Life in Hours of 200 Incandescent Lamps*

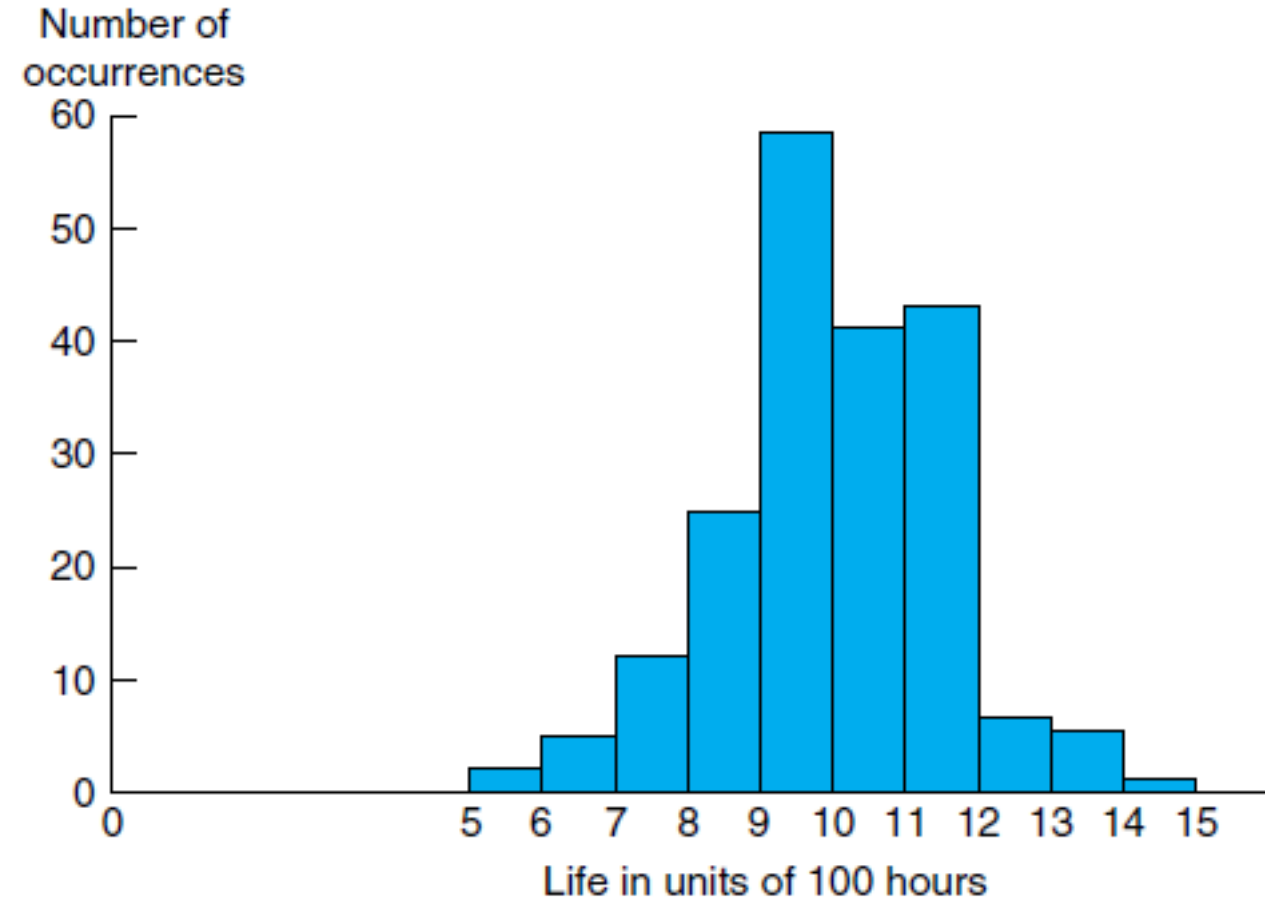
Item Lifetimes									
1,067	919	1,196	785	1,126	936	918	1,156	920	948
855	1,092	1,162	1,170	929	950	905	972	1,035	1,045
1,157	1,195	1,195	1,340	1,122	938	970	1,237	956	1,102
1,022	978	832	1,009	1,157	1,151	1,009	765	958	902
923	1,333	811	1,217	1,085	896	958	1,311	1,037	702
521	933	928	1,153	946	858	1,071	1,069	830	1,063
930	807	954	1,063	1,002	909	1,077	1,021	1,062	1,157
999	932	1,035	944	1,049	940	1,122	1,115	833	1,320
901	1,324	818	1,250	1,203	1,078	890	1,303	1,011	1,102
996	780	900	1,106	704	621	854	1,178	1,138	951
1,187	1,067	1,118	1,037	958	760	1,101	949	992	966
824	653	980	935	878	934	910	1,058	730	980
844	814	1,103	1,000	788	1,143	935	1,069	1,170	1,067
1,037	1,151	863	990	1,035	1,112	931	970	932	904
1,026	1,147	883	867	990	1,258	1,192	922	1,150	1,091
1,039	1,083	1,040	1,289	699	1,083	880	1,029	658	912
1,023	984	856	924	801	1,122	1,292	1,116	880	1,173
1,134	932	938	1,078	1,180	1,106	1,184	954	824	529
998	996	1,133	765	775	1,105	1,081	1,171	705	1,425
610	916	1,001	895	709	860	1,110	1,149	972	1,002



Histogram

TABLE 2.4 A Class Frequency Table

Class Interval	Frequency (Number of Data Values in the Interval)
500–600	2
600–700	5
700–800	12
800–900	25
900–1000	58
1000–1100	41
1100–1200	43
1200–1300	7
1300–1400	6
1400–1500	1



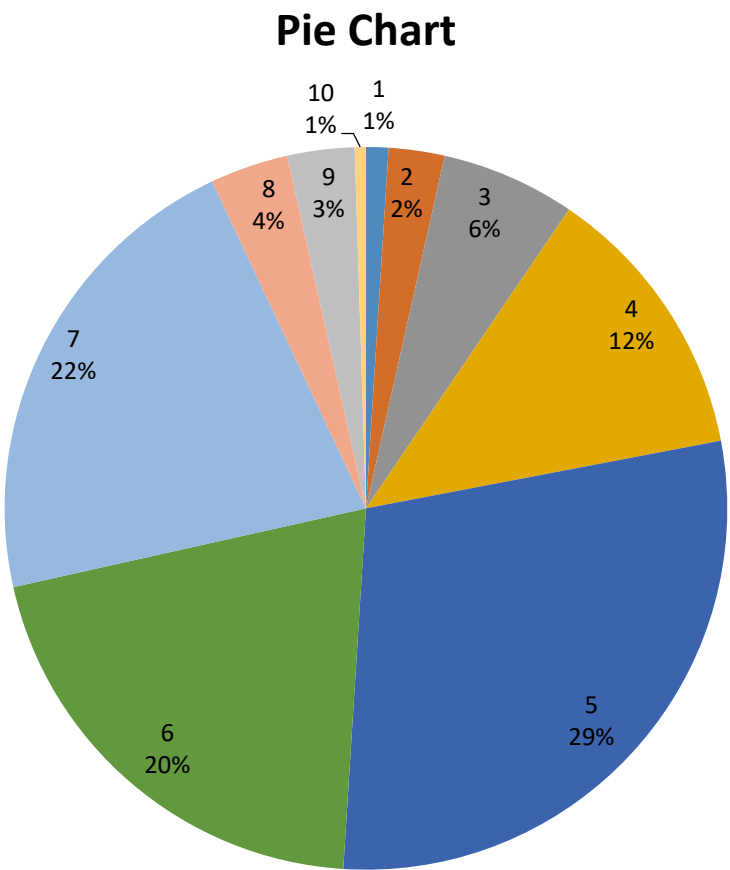
Frequency histogram and Relative frequency histogram

Relative frequency histogram

Sr.No	Class Interval	Frequency	Relative Frequency
1	500-600	2	0.01
2	600-700	5	0.025
3	700-800	12	0.06
4	800-900	25	0.125
5	900-1000	58	0.29
6	1000-1100	41	0.205
7	1100-1200	43	0.215
8	1200-1300	7	0.035
9	1300-1400	6	0.03
10	1400-1500	1	0.005

Relative frequency histogram

Sr.No	Class Interval	Frequency	Relative Frequency
1	500-600	2	0.01
2	600-700	5	0.025
3	700-800	12	0.06
4	800-900	25	0.125
5	900-1000	58	0.29
6	1000-1100	41	0.205
7	1100-1200	43	0.215
8	1200-1300	7	0.035
9	1300-1400	6	0.03
10	1400-1500	1	0.005

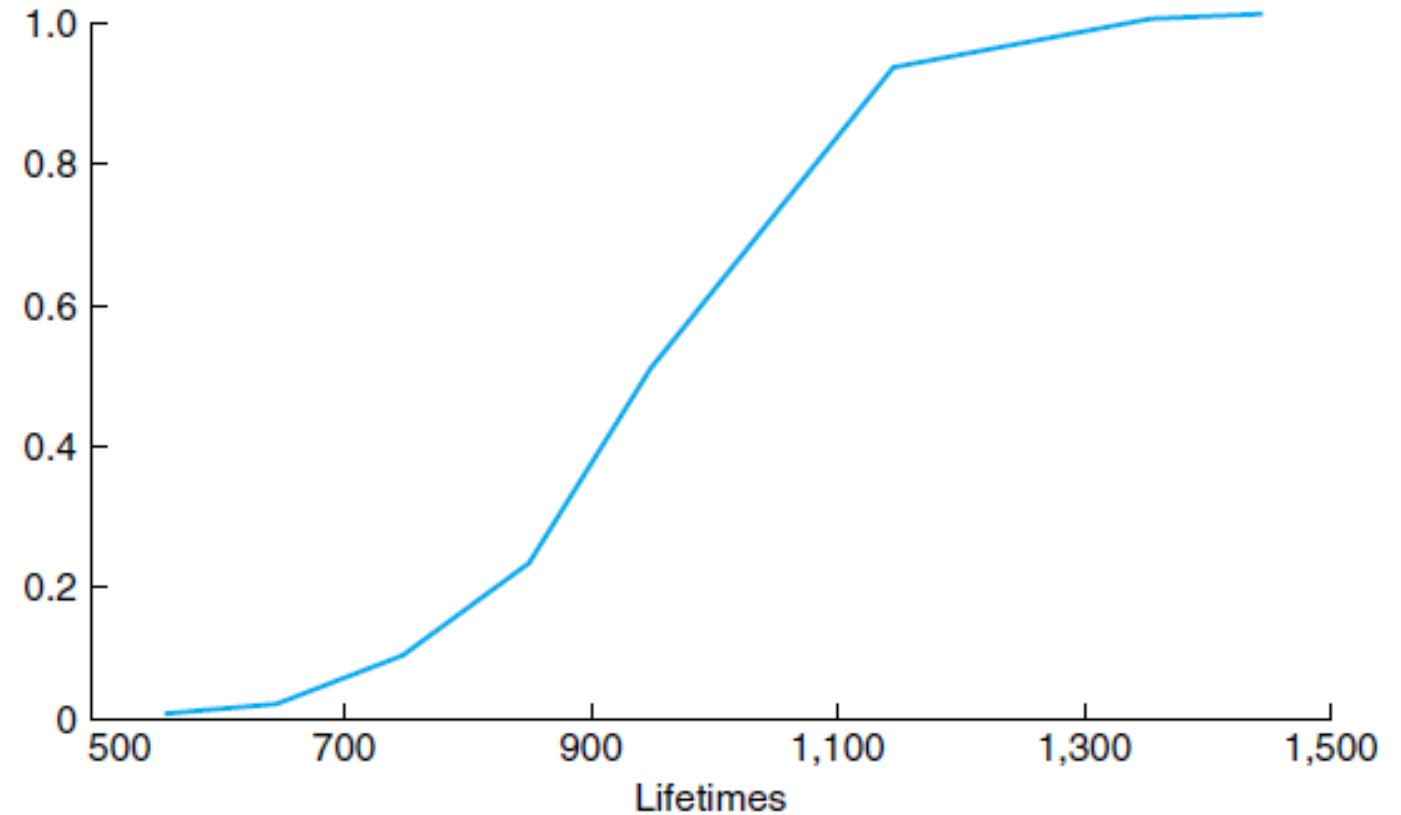


Cumulative frequency histogram

Sr.No	Class Interval	Frequency	Relative Frequency	Cumulative Frequency
1	500-600	2	0.01	0.01
2	600-700	5	0.025	0.035
3	700-800	12	0.06	0.095
4	800-900	25	0.125	0.22
5	900-1000	58	0.29	0.51
6	1000-1100	41	0.205	0.715
7	1100-1200	43	0.215	0.93
8	1200-1300	7	0.035	0.965
9	1300-1400	6	0.03	0.995
10	1400-1500	1	0.005	1

Cumulative frequency histogram: Ogive

Sr.No	Class Interval	Cumulative Frequency
1	500-600	0.01
2	600-700	0.035
3	700-800	0.095
4	800-900	0.22
5	900-1000	0.51
6	1000-1100	0.715
7	1100-1200	0.93
8	1200-1300	0.965
9	1300-1400	0.995
10	1400-1500	1



Stem and leaf plot

- An efficient way of organizing a small- to moderate-sized data set

Thus, for instance, the value 62 is expressed as

Stem Leaf

6 2

and the two data values 62 and 67 can be represented as

Stem Leaf

6 2,7

State	Station	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Annual avg.
AL	Mobile	40.0	42.7	50.1	57.1	64.4	70.7	73.2	72.9	68.7	57.3	49.1	43.1	57.4
AK	Juneau	19.0	22.7	26.7	32.1	38.9	45.0	48.1	47.3	42.9	37.2	27.2	22.6	34.1
AZ	Phoenix	41.2	44.7	48.8	55.3	63.9	72.9	81.0	79.2	72.8	60.8	48.9	41.8	59.3
AR	Little Rock	29.1	33.2	42.2	50.7	59.0	67.4	71.5	69.8	63.5	50.9	41.5	33.1	51.0
CA	Los Angeles	47.8	49.3	50.5	52.8	56.3	59.5	62.8	64.2	63.2	59.2	52.8	47.9	55.5
	Sacramento	37.7	41.4	43.2	45.5	50.3	55.3	58.1	58.0	55.7	50.4	43.4	37.8	48.1
	San Diego	48.9	50.7	52.8	55.6	59.1	61.9	65.7	67.3	65.6	60.9	53.9	48.8	57.6
	San Francisco	41.8	45.0	45.8	47.2	49.7	52.6	53.9	55.0	55.2	51.8	47.1	42.7	49.0
CO	Denver	16.1	20.2	25.8	34.5	43.6	52.4	58.6	56.9	47.6	36.4	25.4	17.4	36.2
CT	Hartford	15.8	18.6	28.1	37.5	47.6	56.9	62.2	60.4	51.8	40.7	32.8	21.3	39.5
DE	Wilmington	22.4	24.8	33.1	41.8	52.2	61.6	67.1	65.9	58.2	45.7	37.0	27.6	44.8
DC	Washington	26.8	29.1	37.7	46.4	56.6	66.5	71.4	70.0	62.5	50.3	41.1	31.7	49.2
FL	Jacksonville	40.5	43.3	49.2	54.9	62.1	69.1	71.9	71.8	69.0	59.3	50.2	43.4	57.1
	Miami	59.2	60.4	64.2	67.8	72.1	75.1	76.2	76.7	75.9	72.1	66.7	61.5	69.0
GA	Atlanta	31.5	34.5	42.5	50.2	58.7	66.2	69.5	69.0	63.5	51.9	42.8	35.0	51.3
HI	Honolulu	65.6	65.4	67.2	68.7	70.3	72.2	73.5	74.2	73.5	72.3	70.3	67.0	70.0
ID	Boise	21.6	27.5	31.9	36.7	43.9	52.1	57.7	56.8	48.2	39.0	31.1	22.5	39.1
IL	Chicago	12.9	17.2	28.5	38.6	47.7	57.5	62.6	61.6	53.9	42.2	31.6	19.1	39.5
	Peoria	13.2	17.7	29.8	40.8	50.9	60.7	65.4	63.1	55.2	43.1	32.5	19.3	41.0
IN	Indianapolis	17.2	20.9	31.9	41.5	51.7	61.0	65.2	62.8	55.6	43.5	34.1	23.2	42.4
IA	Des Moines	10.7	15.6	27.6	40.0	51.5	61.2	66.5	63.6	54.5	42.7	29.9	16.1	40.0
KS	Wichita	19.2	23.7	33.6	44.5	54.3	64.6	69.9	67.9	59.2	46.6	33.9	23.0	45.0
KY	Louisville	23.2	26.5	36.2	45.4	54.7	62.9	67.3	65.8	58.7	45.8	37.3	28.6	46.0
LA	New Orleans	41.8	44.4	51.6	58.4	65.2	70.8	73.1	72.8	69.5	58.7	51.0	44.8	58.5
ME	Portland	11.4	13.5	24.5	34.1	43.4	52.1	58.3	57.1	48.9	38.3	30.4	17.8	35.8
MD	Baltimore	23.4	25.9	34.1	42.5	52.6	61.8	66.8	65.7	58.4	45.9	37.1	28.2	45.2
MA	Boston	21.6	23.0	31.3	40.2	49.8	59.1	65.1	64.0	56.8	46.9	38.3	26.7	43.6
MI	Detroit	15.6	17.6	27.0	36.8	47.1	56.3	61.3	59.6	52.5	40.9	32.2	21.4	39.0
	Sault Ste. Marie	4.6	4.8	15.3	28.4	38.4	45.5	51.3	51.3	44.3	36.2	25.9	11.8	29.8
MN	Duluth	-2.2	2.8	15.7	28.9	39.6	48.5	55.1	53.3	44.5	35.1	21.5	4.9	29.0
	Minneapolis-St. Paul ..	2.8	9.2	22.7	36.2	47.6	57.6	63.1	60.3	50.3	38.8	25.2	10.2	35.3
MS	Jackson	32.7	35.7	44.1	51.9	60.0	67.1	70.5	69.7	63.7	50.3	42.3	36.1	52.0
MO	Kansas City	16.7	21.8	32.6	43.8	53.9	63.1	68.2	65.7	56.9	45.7	33.6	21.9	43.7
	St. Louis	20.8	25.1	35.5	46.4	56.0	65.7	70.4	67.9	60.5	48.3	37.7	26.0	46.7
MT	Great Falls	11.6	17.2	22.8	31.9	40.9	48.6	53.2	52.2	43.5	35.8	24.3	14.6	33.1

Stem and leaf plot

The annual average daily minimum temperatures from Table in previous slide are represented in the following stem and leaf plot.

7	0.0
6	9.0
5	1.0, 1.3, 2.0, 5.5, 7.1, 7.4, 7.6, 8.5, 9.3
4	0.0, 1.0, 2.4, 3.6, 3.7, 4.8, 5.0, 5.2, 6.0, 6.7, 8.1, 9.0, 9.2
3	3.1, 4.1, 5.3, 5.8, 6.2, 9.0, 9.5, 9.5
2	9.0, 9.8

Home work

- **What is Exploratory Data Analysis ?**
- Exploratory Data Analysis or (EDA) is understanding the data sets by summarizing their main characteristics often plotting them visually.
- **How to perform Exploratory Data Analysis ?**
- This is one such question that everyone is keen on knowing the answer. Well, **the answer is it depends on the data set that you are working.** There is no one method or common methods in order to perform EDA, whereas in this tutorial you can understand some common methods and plots that would be used in the EDA process.
- <https://www.kaggle.com/code/imoore/intro-to-exploratory-data-analysis-eda-in-python/notebook>